

Lecture 4: The Capacitor

an ideal energy-storage element — and the first that remembers

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

■ charge ■ current ■ voltage ■ power, energy

An element that stores energy

Lecture 3

a curve in the $v-i$ plane
energy in $\rightarrow$ dissipated as heat



Today

a curve in the $v-q$ plane
energy in $\rightarrow$ stored $\rightarrow$ back out

A resistor **dissipates**: $p = Ri^2 \geq 0$ always, so the energy it is given turns into heat and never comes back.

Today's element dissipates nothing. It is an **ideal energy-storage element** — everything put in can be taken back out, which is also why it **remembers**.

Every domain has one

capacitor

voltage v

$$\frac{1}{2} C v^2$$

moving mass

velocity u

$$\frac{1}{2} m u^2$$

flywheel

angular velocity ω

$$\frac{1}{2} J \omega^2$$

Charge is electrical bookkeeping. **Energy** is what every domain shares — Lecture 1's table, read one level higher.

Each holds half a constant times the square of one variable, and gives all of it back.

The resistor's mechanical counterpart is *viscous friction*.

Some elements store charge

For those, the useful pair is not (v, i) but (v, q) .

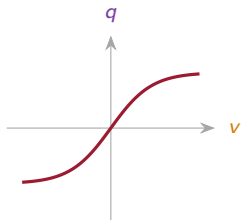
$$q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau \quad \Longleftrightarrow \quad i = \frac{dq}{dt}$$

q is the charge that has entered the $+$ terminal.

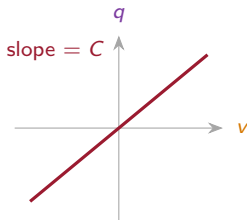
The link between the two is Lecture 1, unchanged.

What a capacitor is

A two-terminal element is a **capacitor** if its (v, q) pairs lie on a **curve in the v - q plane**.



in general



LTI: a straight line

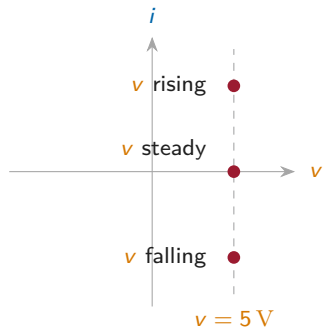
A bent curve is a nonlinear capacitor — a varactor diode is one.

In this course every capacitor is **linear and time-invariant**: from here on, the straight line.

And note what has *not* happened

A capacitor has **no characteristic** in the v - i plane.

There *is* a relation between v and i — but not between their values at one instant.

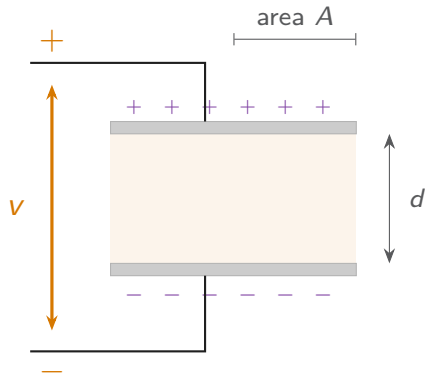


No f , no g :

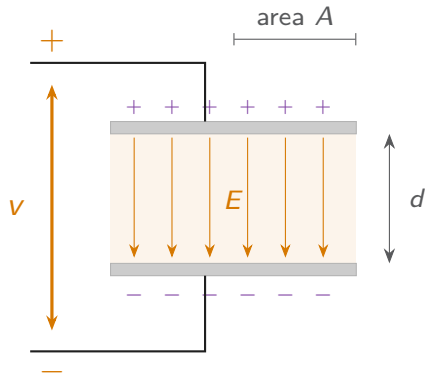
$$i(t) \neq f(v(t)), \quad v(t) \neq g(i(t))$$

One voltage, **any** current — it follows how v is *changing*, not what it is.

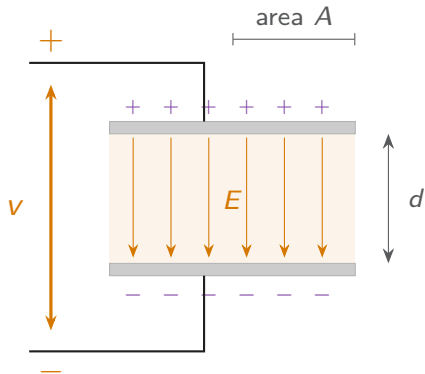
Where the charge sits



Where the charge sits



Where the charge sits



$$q = C v, \quad C = \frac{\epsilon A}{d}$$

Unit: the **farad**, $F = C/V$. Wide plates, close together, with a high-permittivity material between them.

One farad is enormous

1 cm² of plate, 0.1 mm of air:

$$C = \frac{\epsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(10^{-4})}{10^{-4}} \approx 8.9 \text{ pF}$$

Which is why real values come in **pF**, **nF** and **μF** —
and why a 1 F supercapacitor has to roll an enormous area into a small can.

The LTI capacitor

$q = C v$ with C constant. Differentiate:

$$i = C \frac{dv}{dt}$$

differential

$$v(t) = v(t_0) + \frac{1}{C} \int_{t_0}^t i(\tau) d\tau$$

integral

The current is whatever the **rate of change** of voltage demands.
The voltage is the **running total** of everything the current has done.

Four consequences

Open circuit at DC	Voltage cannot jump	It remembers	It holds energy
v constant $\Rightarrow$ $dv/dt = 0 \Rightarrow i = 0$.	Not with a finite current — a step needs $dv/dt = \infty$.	$v(t)$ depends on the <i>whole history</i> of i .	$w = \frac{1}{2} C v^2$, set by the present voltage alone.

A resistor knows only *now*, and keeps nothing.

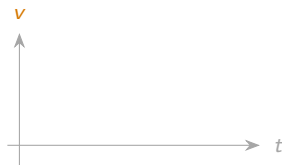
A capacitor knows everything that has happened to it — and **hands the energy back**.

Example: push a constant current in

$$I = 1 \text{ mA into } C = 1 \mu\text{F, from } v(0) = 0$$



current in



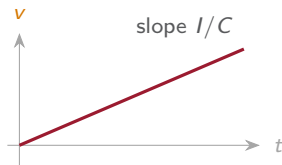
voltage ramps

Example: push a constant current in

$$I = 1 \text{ mA into } C = 1 \mu\text{F, from } v(0) = 0 \quad v(t) = \frac{I}{C} t = 1000 t \text{ [V]}$$



current in



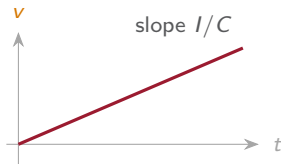
voltage ramps

Example: push a constant current in

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current in



voltage ramps

The capacitor is **integrating** the current.

5 V after 5 ms — how a saw-tooth generator works.

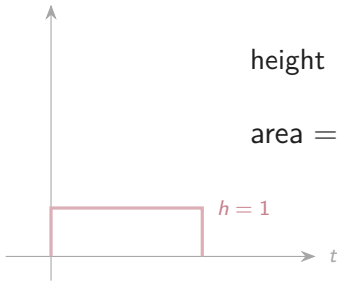
Example: hold the voltage constant

$$v = V \text{ constant} \implies i = C \frac{dv}{dt} = 0, \quad q = CV \text{ sits there}$$

The capacitor has become an open circuit, storing CV while it does so.

But how did the voltage get there?

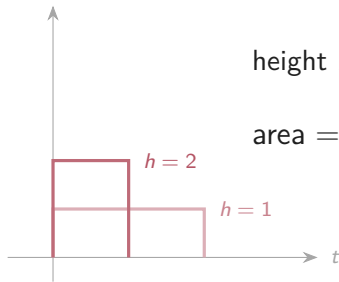
A signal we have not met: the impulse



height h , width $1/h$

$$\text{area} = h \cdot \frac{1}{h} = 1 \text{ always}$$

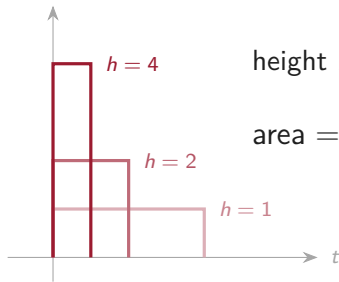
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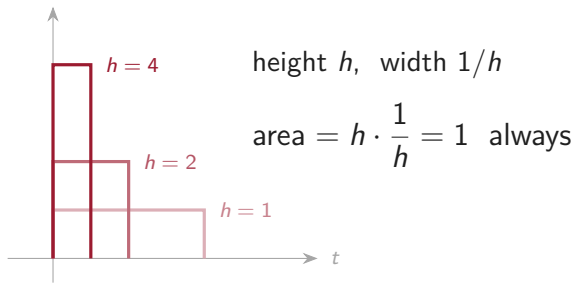
A signal we have not met: the impulse



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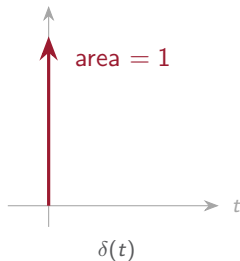
A signal we have not met: the impulse



Let h grow: taller, narrower, **the same area**.

In the limit the width goes to zero and no value is left — only the area.

The unit impulse $\delta(t)$



$$\delta(t) = 0 \text{ for } t \neq 0, \quad \int_a^b \delta(\tau) d\tau = 1$$

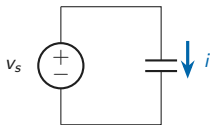
for any $a < 0 < b$

Not a function — it is defined by its **area**.
The number beside the arrow is its area, **not a height**.

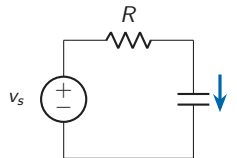
$K \delta(t)$ has area K .

Two uses today: a current impulse $K\delta(t)$ delivers **charge K** in zero time, and
if **v** jumps by V at $t = 0$ then $\frac{dv}{dt} = V \delta(t)$.

The source steps from 0 to V . What is the current?



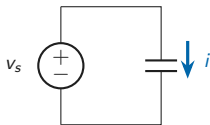
nothing in series



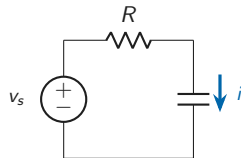
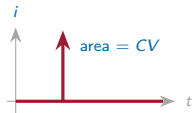
any real circuit



The source steps from 0 to V . What is the current?



nothing in series

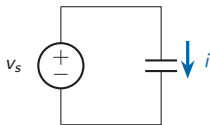


any real circuit

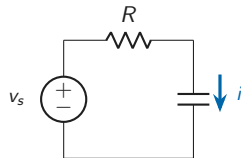
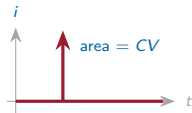


Differentiate the step: $i = C dv/dt = CV \delta(t)$ — an **impulse** of area CV .

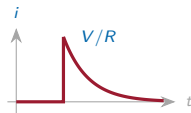
The source steps from 0 to V . What is the current?



nothing in series



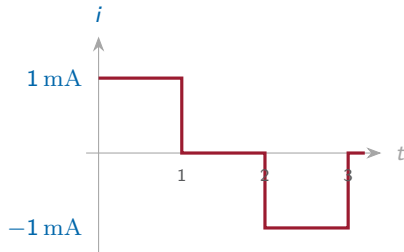
any real circuit



Differentiate the step: $i = C dv/dt = CV \delta(t)$ — an **impulse** of area CV .

With R there the current starts at V/R and decays while v rises smoothly to V — we will solve this circuit in detail later.

Example: read a current waveform

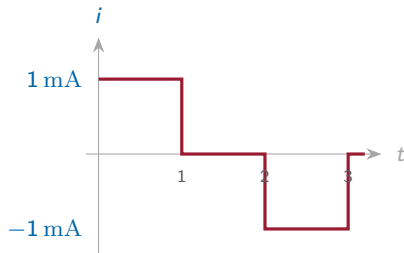


current in, ms

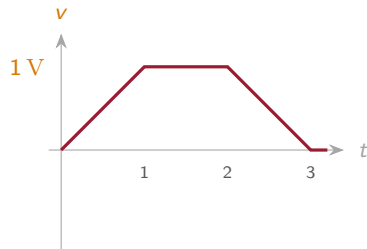


voltage out, ms

Example: read a current waveform



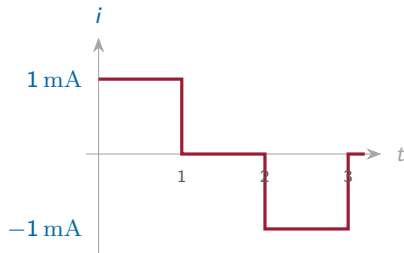
current in, ms



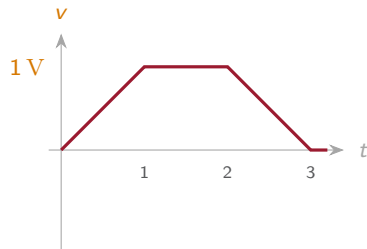
voltage out, ms

Constant current $\Rightarrow$ straight line, slope i/C . Zero current $\Rightarrow$ the voltage holds.

Example: read a current waveform



current in, ms



voltage out, ms

Constant current $\Rightarrow$ straight line, slope i/C . Zero current $\Rightarrow$ the voltage holds.

It ends where it started, having given back exactly the charge it took.

Energy stored in a capacitor

$$p = v i = v C \frac{dv}{dt} = \frac{d}{dt} \left(\frac{1}{2} C v^2 \right)$$

The power is the **rate of change** of $\frac{1}{2} C v^2$.
So that is the energy the capacitor holds:

$$w = \frac{1}{2} C v^2 = \frac{q^2}{2C}$$

It depends only on the **present** voltage — not on how the capacitor got there.

Resistor and capacitor, side by side

	LTI resistor (Lecture 3)	LTI capacitor
terminal equation	$v = Ri$	$i = C dv/dt$
plane it lives in	$v-i$	$v-q$
memory	none	the whole past of i
power	$Ri^2 \geq 0$ always	either sign
energy	dissipated as heat, gone	stored, all of it returnable
verdict	passive, lossy	passive, lossless

For the mechanically minded

$$i = C \frac{d\boldsymbol{v}}{dt} \quad \longleftrightarrow \quad F = m \frac{du}{dt}$$

electrical		mechanical
current i	$\leftrightarrow$	force F
voltage $\boldsymbol{v}$	$\leftrightarrow$	velocity u
capacitance C	$\leftrightarrow$	mass m
$\frac{1}{2} C \boldsymbol{v}^2$	$\leftrightarrow$	$\frac{1}{2} m u^2$

A heavy flywheel is a large capacitance.

A finite force cannot change its speed instantly —
and a finite current cannot change a capacitor's voltage instantly.

Summary

- An **ideal energy-storage element**: it dissipates nothing, and everything put in can be taken back out. Lecture 3's elements dissipated all of it.
- A **capacitor** is a curve in the v - q plane, not a resistor — no f with $i(t) = f(v(t))$.
- $C = \epsilon A/d$, in farads. One farad is enormous; real values are pF, nF, μ F.
- **LTI capacitor**: $i = C dv/dt$, or the integral form. Open circuit at DC; voltage cannot jump; it remembers.
- $w = \frac{1}{2} C v^2 = q^2/2C$ — stored, not spent; set by the present voltage alone. Passive and **lossless**.

Next lecture

The inductor

The *second* ideal energy-storage element, and the mirror image of today: flux linkage and current in place of charge and voltage, $\frac{1}{2}Li^2$ in place of $\frac{1}{2}Cv^2$.

Questions?

Notes for this lecture: see the course repository.